Teymour Gray

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Education

PhD, University College London Sept 2023 – Sept 2027 (expected)

Supervised by Luis García Martinez, funded by 4-year Teaching Assistant scholarship. Research area algebraic number theory. PhD thesis 'Stark's Conjecture and the Cohomology of Arithmetic Groups' expected September 2027.

MSci Mathematics, University College London Sept 2019 – June 2023

First-Class Honours. Dissertation on Hauptmoduls of genus zero modular curves. Awarded IMA Graduate Prize & David G. Larman Prize for best performance in Pure Mathematics.

Work Experience

Teaching Assistant at University College London Sept 2023 – Present

Publications

Cocycle Relations and Elliptic Gamma Values, Pre-print, October 2026 (arXiv).

Selected Talks

Arithmetic and Geometric Reciprocity Laws KCL, Nov 2026

Quantum dilogarithms: Stark units for real quadratic fields ICL, Nov 2026

Stark's Conjecture and Elliptic Units Oxford Junior Number Theory Seminar, May 2026

Global Class Field Theory: the Ambiguous Class Number Formula London Number Theory Study Group, Feb 2026

Stark's Conjecture and Elliptic Units London Junior Number Theory Seminar, Nov 2025

Why Algebraic Geometry? How geometry can illuminate algebra UCL, July 2025

Special Values of L-functions: an algebraic approach to rationality UCL, June 2025

Iwasawa Theory: proof of the IMC KCL, June 2025

The Iwasawa algebra, cyclotomic units and the Coleman map KCL, May 2025

On the structure of adjoint Chevalley groups KCL, Nov 2024

Conferences and Summer Schools

[TBC] Quantum Designs, Stark Units and Hilbert's 12th Problem: Conference Bernoulli Center, June 2027

[TBC] Explicit Class Field Theory: Arbeitsgemeinschaft Oberwolfach, March 2027

p-adic aspects of the Kudla program: Conference CIRM, June 2026

Classical and p-adic aspects of the Kudla program: Research School CIRM, June 2026

The Theta Correspondence: Research School UEA, July 2025

Eisenstein Days: Workshop on zeta values and Eisenstein cocycles UCL, Dec 2023

Teaching Experience

Course Leader for Differential and Integral Calculus Sept 2024 – Sept 2026

Taught undergraduate course (MATH0039) for two years with E. Atlason. Course planning, lecture delivery, and preparation of lecture notes, problem sheets and exams. Awarded Mathematical & Physical Sciences Faculty Education Award for 2026.

Generating Functions Course May 2024

Designed and taught a four-week course on generating functions for sixth-formers.

Category Theory Lecture Series March 2024

Designed and taught a three-part lecture series for fourth-year UCL students. I covered functors, natural transformations, the Yoneda Lemma, and categorical (co)-limits.

Tutorial Leader Sept 2023 – Present

Tutored Algebra 2 (Groups & Rings) and Algebra 3 (Further Linear Algebra).

Teaching Talks

What's the difference between a maths lecture and a joke? Academic Teaching Workshop KCL, June 2026

Experiences from delivering a lecture course UCL/KCL Mathematics and Statistics Education Seminar, Dec 2025

How to teach a lecture course ECR Seminar UCL, Oct 2025

Organisation

London Maths Outreach Founding Committee Member Dec 2023 – Present

- Developed the LMO program: two four-week courses every academic half-term, introducing state-school students (aged 14-18) to university-style mathematics, with an emphasis on exploration.
- Steered LMO's application for charitable status.
- Coordinate communications with schools, students and parents.
- Organise further outreach initiatives, including our annual Summer Day and in-schools Workshop Program.

LMO received a 'President's Award for Excellence in Societal Engagement' from Imperial College London in 2026.

Outreach

LMO Summer Day Aug 2024 – Present

Co-organised annual LMO Summer Day in 2024, 2025, and 2026. Features mathematical workshops, invited talks and team challenges for school students. The previous two events attracted approximately 60 attendees.

LMO Workshop Program July 2026 – Present

Established the LMO workshop program in 2026, delivering mathematical workshops at schools and outreach events. So far, I have presented workshops to over 200 students. Multiple sessions arranged with schools in the coming academic year.

Awards

Mathematical & Physical Sciences Faculty Education Award 2026

Institute of Mathematics and its Applications Graduate Prize 2023

David G. Larman Prize*: top fourth-year MSci student in Pure Mathematics* 2023

Dean's List Commendation*: top 5% in Mathematical & Physical Sciences* 2023

Nazir Ahmad Prize*: top third-year MSci student in Mathematics* 2022

Jackson Lewis Commendation 2022

Referees

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